Lab 7 – Fuzzing for Input Validation Bugs

Project: customs_duty.py (parse_declaration, duty_amount, hs_chapter) – Hypothesis 6.170.15, pytest 9.1.1, Python 3.13

Step 1 – Install dependencies

```
$ pip install hypothesis pytest
Successfully installed hypothesis-6.170.15 (pytest 9.1.1 already present)
Python 3.13.18
```

Step 2 – Run the original (buggy) script

```
$ python customs_duty.py
```

Insert your own screenshot of running the original **customs_duty.py** here (not captured by the assistant).

Step 4 – Run test_customs_duty.py against the ORIGINAL code (failing)

```
$ pytest test_customs_duty.py -v
============================= test session starts ==============================
platform linux -- Python 3.13.18, pytest-9.1.1, pluggy-1.6.0 -- /usr/local/bin/python3
hypothesis profile 'default'
rootdir: /tmp/claude-0/-home-claude/c550fd4c-5834-5672-b5e3-f9d0000edd64/scratchpad/before
plugins: platformdirs-4.13.5, anyio-4.15.1, hypothesis-6.170.15
collecting ... collected 14 items

test_customs_duty.py::test_parse_declaration_no_crash FAILED             [  7%]
test_customs_duty.py::test_duty_amount_no_crash FAILED                   [ 14%]
test_customs_duty.py::test_hs_chapter_no_crash FAILED                    [ 21%]
test_customs_duty.py::test_parse_declaration_any_type_no_crash FAILED    [ 28%]
test_customs_duty.py::test_duty_amount_any_type_no_crash FAILED          [ 35%]
test_customs_duty.py::test_hs_chapter_any_type_no_crash FAILED           [ 42%]
test_customs_duty.py::test_parse_declaration_output_is_valid_or_none FAILED [ 50%]
test_customs_duty.py::test_duty_amount_output_is_valid_or_none FAILED    [ 57%]
test_customs_duty.py::test_hs_chapter_output_is_valid_or_none FAILED     [ 64%]
test_customs_duty.py::test_parse_declaration_roundtrip PASSED            [ 71%]
test_customs_duty.py::test_duty_amount_matches_formula PASSED            [ 78%]
test_customs_duty.py::test_hs_chapter_valid_codes PASSED                 [ 85%]
test_customs_duty.py::test_known_bad_inputs_are_rejected FAILED          [ 92%]
test_customs_duty.py::test_known_good_inputs FAILED                      [100%]

=========================== short test summary info ============================
FAILED test_customs_duty.py::test_parse_declaration_no_crash - ExceptionGroup...
FAILED test_customs_duty.py::test_duty_amount_no_crash - ExceptionGroup: Hypo...
FAILED test_customs_duty.py::test_hs_chapter_no_crash - ExceptionGroup: Hypot...
FAILED test_customs_duty.py::test_parse_declaration_any_type_no_crash - Excep...
FAILED test_customs_duty.py::test_duty_amount_any_type_no_crash - ExceptionGr...
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FAILED test_customs_duty.py::test_duty_amount_output_is_valid_or_none - Excep...
FAILED test_customs_duty.py::test_hs_chapter_output_is_valid_or_none - ValueE...
FAILED test_customs_duty.py::test_known_bad_inputs_are_rejected - AttributeEr...
FAILED test_customs_duty.py::test_known_good_inputs - AssertionError: assert ...
======================== 11 failed, 3 passed in 0.30s ==========================

---- Minimal failing inputs found by Hypothesis (shrunk) ----
    |     assert False, f"duty_amount crashed with: {e!r} for input {s!r}"
    |     assert False, f"hs_chapter crashed with: {e!r} for input {s!r}"
    |     assert False, f"parse_declaration crashed with: {e!r} for input {s!r}"
duty_amount crashed with: AttributeError("'NoneType' object has no attribute 'split'") for input None
duty_amount crashed with: TypeError("a bytes-like object is required, not 'str'") for input b''
duty_amount crashed with: ValueError("could not convert string to float: ''") for input '@'
duty_amount crashed with: ValueError('not enough values to unpack (expected 2, got 1)') for input ''
hs_chapter crashed with: AttributeError("'NoneType' object has no attribute 'split'") for input None
hs_chapter crashed with: TypeError("a bytes-like object is required, not 'str'") for input b''
hs_chapter crashed with: ValueError("invalid literal for int() with base 10: ''") for input ''
parse_declaration crashed with: AttributeError("'NoneType' object has no attribute 'split'") for input None
parse_declaration crashed with: TypeError("a bytes-like object is required, not 'str'") for input b''
parse_declaration crashed with: ValueError("could not convert string to float: ''") for input ':'
parse_declaration crashed with: ValueError('not enough values to unpack (expected 2, got 1)') for input ''
```

Step 5 – Run the FIXED script

```
$ python customs_duty.py
==== Running parse_declaration ====
Enter category:value (e.g. electronics:25000): Category and Value: ('electronics', 25000.0)

==== Running duty_amount ====
Enter value@rate (e.g. 25000@20): Duty Amount: 5000.0

==== Running hs_chapter ====
Enter an HS code (e.g. 8517.12): Chapter: 85

--- invalid inputs ---
==== Running parse_declaration ====
Enter category:value (e.g. electronics:25000): Category and Value: Invalid input

==== Running duty_amount ====
Enter value@rate (e.g. 25000@20): Duty Amount: Invalid input

==== Running hs_chapter ====
Enter an HS code (e.g. 8517.12): Chapter: Invalid input
```

Step 6 – Re-run test_customs_duty.py against the FIXED code (passing)

```
$ pytest test_customs_duty.py -v
============================= test session starts ==============================
platform linux -- Python 3.13.18, pytest-9.1.1, pluggy-1.6.0 -- /usr/local/bin/python3
hypothesis profile 'default'
rootdir: /home/claude/work/L6_L7_35_customs_duty/lab7
plugins: platformdirs-4.13.5, anyio-4.15.1, hypothesis-6.170.15
collecting ... collected 14 items

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test_customs_duty.py::test_known_good_inputs PASSED                      [100%]

============================== 14 passed in 0.46s ==============================
```

Bugs identified and fixed

Function	Fuzz input	Original behaviour	Fix
all three	None, bytes	AttributeError / TypeError	isinstance(str) guard, return None
parse_declaration	'' / 'a:1:2'	ValueError (unpack)	require exactly one ':'
parse_declaration	':' , 'a:'	ValueError float('')	validate number and non-empty category
parse_declaration	'a:nan', 'a:inf', 'a:-5'	accepted silently	finite, non-negative values only (ASCII regex)
duty_amount	'', '@', '5@5@5'	ValueError	require exactly one '@', validate both numbers
duty_amount	'5@nan', '-5@10', '1e308@1e308'	nan / negative / inf duty	reject; result must be finite
hs_chapter	'', '.', 'ab.12'	ValueError int('')	need 2 ASCII digits
hs_chapter	'5', '00.1', '98.1', '-1.2', Arabic-Indic digits	wrong chapter or accepted	chapter limited to 1–97, ASCII digits only

hs_chapter	' 0101.21 '	returned 0 (whitespace misparsed)	strip input first